Geometrically, $x = 0$ namely the y-axis is a vertical asymptote to the curve representing $f(x) = \frac{1}{x^2}$.

The graph of the function $f(x) = \frac{1}{x^2}$ is shown in Fig. 9.22.

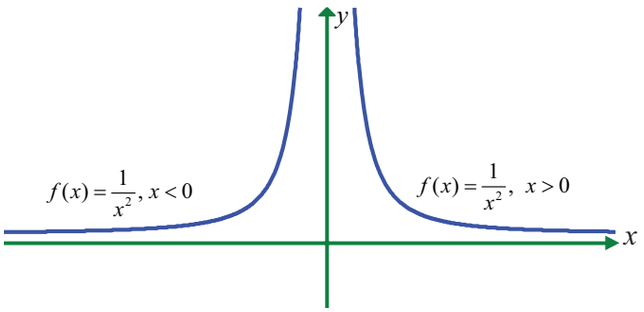


Fig. 9.22

Remember that the limit is infinite and so $\lim_{x \to 0} \frac{1}{x^2}$ does not exist. Students are cautioned that ∞ is a symbol for this behaviour of $f(x) = \frac{1}{x^2}$ and is not a new number.

Similarly, if we look at $f(x) = \frac{1}{x}$,

it is easy to see that, $\frac{1}{x} \to -\infty$ as $x \to 0^-$

and $\frac{1}{x} \to +\infty$ as $x \to 0^+$

which is geometrically clear from the graph of $f(x) = \frac{1}{x}$ (Fig. 9.23) .

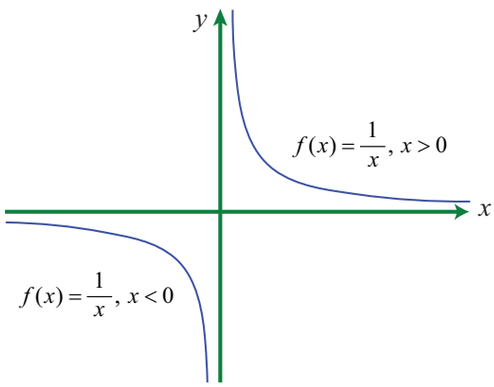


Fig. 9.23

In general we have the following intuitive definitions.

Definition 9.4

For a given $M > 0$, open intervals of the form (M, ∞) is called neighbourhood of ∞.

Similarly, for given $K < 0$, open intervals of the form $(-\infty, K)$ is called neighbourhood of $-\infty$.

Definition 9.5

We say, $f(x) \to \infty$ as x approaches x_0 if for given positive number M there is a neighbourhood of x_0, such that whenever x is in the neighbourhood of $x_0, f(x) > M$. i.e., $f(x) \in (M, \infty)$.

Similarly, $f(x) \to -\infty$ as x approaches x_0 if for a given $K < 0$ there is a neighbourhood of x_0 such that whenever x is in the neighbourhood of $x_0, f(x) < K$. i.e., $f(x) \in (-\infty, K)$.

To describe this situation symbolically, we write

$f(x) \to \infty$ as $x \to x_0$

$f(x) \to -\infty$ as $x \to x_0$

$f(x) \to \infty$ as $x \to x_0^-$

$f(x) \to -\infty$ as $x \to x_0^+$

and $f(x) \to \infty$ as $x \to x_0^+$

$f(x) \to -\infty$ as $x \to x_0^-$

are called **infinite limits**. If any one of the foregoing conditions hold, then the line $x = x_0$ is a **vertical asymptote** for the graph of $f(x)$.

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